COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-007: Wireless Site Survey

Document ID	TS-INF-007
Title	Wireless Site Survey
Executive Sponsor	OIT Network Enterprise Engineering
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT staff, OIT-supported agencies, and contractors, vendors, and integrators. Governs all net-new and go-forward wireless deployments; not retroactively enforceable for legacy systems unless part of modernization.
Tier	Standard
Version	4.0
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1. Purpose

In accordance with C.R.S. §24-37.5, the Office of Information Technology (OIT) is responsible for delivering secure, reliable, and efficient IT services to the State of Colorado and its supported agencies. The purpose of this standard is to define mandatory requirements and engineering guidance for planning, validating, and documenting wireless network deployments under the OIT Enterprise Wireless Solution.

Wireless Site Surveys ensure: wireless networks are designed based on measured RF conditions; Access Point (AP) quantity, type, and placement support coverage, capacity, and performance; deployments meet operational, security, and environmental constraints; and post-deployment performance aligns with intended service levels. Both pre-implementation and post-implementation validation surveys are required.

2. Scope

This standard applies to OIT staff, OIT-supported agencies, and contractors, vendors, and integrators. It governs all net-new and go-forward wireless deployments and is not retroactively enforceable for legacy systems unless part of modernization.

3. Roles and Responsibilities

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

OIT - Network Enterprise Engineering

- Owner of Standard; responsible for defining and updating the Enterprise wireless standards.
- Subject matter expert in wireless design and architecture.
- Confers with security teams on State security policies.
- Updates the standards to keep up with new wireless technologies, industry trends, and best practices.

OIT Staff and Contractors

- All personnel engaged in wireless planning or deployment must be familiar with this standard.
- Apply it during design and deployment, and document compliance or approved exceptions.

4. Standards

Required Survey Types

- Budgetary planning: Predictive survey.
- Pre-install validation: Active survey (required).
- Post-install validation: Active survey (required).
- Security analysis: Passive survey (supplemental).
- Active surveys are mandatory for all enterprise deployments to ensure real RF performance is measured.

RF Design Standards

- Frequency Bands: 2.4 GHz and 5 GHz required; 6 GHz included where supported.
- Channelization: 20 MHz baseline; wider channels require justification.
- Coverage — Data: ≥ -67 dBm / ≥ 25 dB SNR. Voice: ≥ -65 dBm / ≥ 30 dB SNR.
- Surveys must identify non-Wi-Fi RF sources, neighboring WLAN interference, and channel overlap/co-channel contention risks.

Office / Administrative Facilities

- Coverage Targets: Signal strength ≥ -67 dBm; SNR ≥ 25 dB; 10–15% cell overlap; packet loss < 2%; latency < 200 ms.
- Conference rooms and training areas must be designed for capacity, not just coverage.
- Band Prioritization: 5 GHz primary; 6 GHz where supported; 2.4 GHz limited to legacy or IoT devices.

Healthcare / Hospital Facilities

- Mission-critical, high-interference, device-dense environments.
- Signal strength ≥ -65 dBm; SNR ≥ 30 dB; 15–20% overlap; packet loss < 1%.
- Consider wet walls, lead shielding, roaming corridors, telemetry, and patient-safe mounting.

Warehouse / Industrial Facilities

- RF-hostile, reflective, and dynamic environments.
- Signal strength ≥ -67 dBm; SNR ≥ 25 dB; retries < 15%.
- Account for metal racking, reflections, high ceilings, moving inventory, and directional antenna needs.

Post-Installation Acceptance Criteria

- Coverage: All areas meet signal thresholds; no dead zones.
- Performance: Roaming validated; throughput meets design.
- RF Health: Acceptable noise floor; minimal co-channel interference.
- Infrastructure: APs installed per design; cabling labeled and tested.
- Security: SSIDs correct; encryption verified; rogue scan complete.
- Documentation: Final maps; heatmaps; validation report; variances documented.

5. Variance and Exception Process

Deviations require a formal Exception Request with business/technical justification, approved by the Network Services Manager.

6. References

- C.R.S. §24-37.5
- OIT Enterprise Wireless Standards
- TS-INF-009 Infrastructure as Code Standard

7. Definitions

Active Survey: RF survey where a client associates with APs to measure performance, data rates, and roaming.

Passive Survey: Listen-only RF scan used for interference or rogue detection.

Predictive Survey: Software modeling based on floor plans without field measurements.

Wireless Access Point (AP): Device providing Wi-Fi connectivity to clients.

Client Device: Laptop, phone, scanner, medical device, or IoT device using Wi-Fi.

Wi-Fi: WLAN technology based on IEEE 802.11 standards.